

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: N,N-Dimethylacetamide  
 Product Description: N,N-Dimethylacetamide  
 Cat No. : C37523  
 CAS No 127-19-5  
 Molecular Formula C4 H9 N O

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                    |                     |
|------------------------------------|---------------------|
| Acute dermal toxicity              | Category 4 (H312)   |
| Acute Inhalation Toxicity - Vapors | Category 4 (H332)   |
| Serious Eye Damage/Eye Irritation  | Category 2 (H319)   |
| Reproductive Toxicity              | Category 1B (H360D) |

**Label Elements**


Signal Word

Danger

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## Hazard Statements

H312 + H332 - Harmful in contact with skin or if inhaled  
H319 - Causes serious eye irritation  
H360D - May damage the unborn child

## Precautionary Statements

### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear eye protection/ face protection

### Response

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 + P235 - Store in a well-ventilated place. Keep cool

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Combustible liquid

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component          | CAS No   | Weight % |
|--------------------|----------|----------|
| Dimethyl acetamide | 127-19-5 | >95      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a

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pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Most important symptoms and effects, both acute and delayed**

None reasonably foreseeable. Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

**Extinguishing media**

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Special hazards arising from the substance or mixture**

Combustible material. Containers may explode when heated. Keep product and empty container away from heat and sources of ignition. Thermal decomposition can lead to release of irritating gases and vapors.

**Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>).

**Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods and Material for Containment and Cleaning Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

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## Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

## Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Keep under nitrogen. Store under an inert atmosphere. Keep container tightly closed in a dry and well-ventilated place. Protect from moisture.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component          | Malaysia | ACGIH TLV           | OSHA PEL                                                                                                         |
|--------------------|----------|---------------------|------------------------------------------------------------------------------------------------------------------|
| Dimethyl acetamide |          | TWA: 10 ppm<br>Skin | (Vacated) TWA: 10 ppm<br>(Vacated) TWA: 35 mg/m <sup>3</sup><br>Skin<br>TWA: 10 ppm<br>TWA: 35 mg/m <sup>3</sup> |

| Component          | European Union                                                                                                                                                                       | The United Kingdom                                                                                                     | Germany                                                                                                                                                                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dimethyl acetamide | TWA: 10 ppm (8h)<br>TWA: 36 mg/m <sup>3</sup> (8h)<br>STEL: 20 ppm (15min)<br>STEL: 72 mg/m <sup>3</sup> (15min)<br>Skin<br><br>STEL: 72 mg/m <sup>3</sup> (8h)<br>STEL: 20 ppm (8h) | STEL: 20 ppm 15 min<br>STEL: 72 mg/m <sup>3</sup> 15 min<br>TWA: 10 ppm 8 hr<br>TWA: 36 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 5 ppm (8 Stunden). AGW -<br>exposure factor 2<br>TWA: 18 mg/m <sup>3</sup> (8 Stunden). AGW -<br>exposure factor 2<br>TWA: 5 ppm (8 Stunden). MAK<br>TWA: 18 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 10 ppm<br>Höhepunkt: 36 mg/m <sup>3</sup><br>Haut |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Goggles

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

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**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice

**Environmental exposure controls** No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                                |                                                 |                                          |
|------------------------------------------------|-------------------------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Colorless                                       |                                          |
| <b>Physical State</b>                          | Liquid                                          |                                          |
| <b>Odor</b>                                    | Ammonia-like                                    |                                          |
| <b>Odor Threshold</b>                          | No data available                               |                                          |
| <b>pH</b>                                      | 4                                               | 200 g/l aq. sol                          |
| <b>Melting Point/Range</b>                     | -20 °C / -4 °F                                  |                                          |
| <b>Softening Point</b>                         | No data available                               |                                          |
| <b>Boiling Point/Range</b>                     | 164 - 166 °C / 327.2 - 330.8 °F                 | @ 760 mmHg                               |
| <b>Flash Point</b>                             | 70 °C / 158 °F                                  | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | <0.17 (Butyl Acetate = 1.0)                     |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable                                  | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.7 vol%<br><b>Upper</b> 11.5 vol% |                                          |
| <b>Vapor Pressure</b>                          | 1.7 mbar @ 25 °C                                |                                          |
| <b>Vapor Density</b>                           | 3.02                                            | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 0.937                                           |                                          |
| <b>Bulk Density</b>                            | Not applicable                                  | Liquid                                   |
| <b>Water Solubility</b>                        | Soluble                                         |                                          |
| <b>Solubility in other solvents</b>            | No information available                        |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                 |                                          |
| <b>Component</b>                               | <b>log Pow</b>                                  |                                          |
| Dimethyl acetamide                             | 0.8                                             |                                          |
| <b>Autoignition Temperature</b>                | 490 °C / 914 °F                                 |                                          |
| <b>Decomposition Temperature</b>               | No data available                               |                                          |
| <b>Viscosity</b>                               | 1.02 mPa s @ 20 °C                              |                                          |
| <b>Explosive Properties</b>                    |                                                 | explosive air/vapour mixtures possible   |
| <b>Oxidizing Properties</b>                    | No information available                        |                                          |
| <b>Molecular Formula</b>                       | C4 H9 N O                                       |                                          |
| <b>Molecular Weight</b>                        | 87.12                                           |                                          |

## SECTION 10: STABILITY AND REACTIVITY

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## Reactivity

None known, based on information available.

## Chemical Stability

Stable under normal conditions. Hygroscopic.

## Possibility of Hazardous Reactions

### **Hazardous Polymerization Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

## Conditions to Avoid

Incompatible products. Heat, flames and sparks. Exposure to moisture. Keep away from open flames, hot surfaces and sources of ignition. Exposure to moist air or water.

## Incompatible Materials

Strong oxidizing agents. Aldehydes. Peroxides. Strong acids.

## Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### **Product Information**

##### **(a) acute toxicity;**

|                   |                   |
|-------------------|-------------------|
| <b>Oral</b>       | No data available |
| <b>Dermal</b>     | No data available |
| <b>Inhalation</b> | No data available |

| Component          | LD50 Oral                 | LD50 Dermal                            | LC50 Inhalation              |
|--------------------|---------------------------|----------------------------------------|------------------------------|
| Dimethyl acetamide | LD50 = 4263 mg/kg ( Rat ) | LD50 = 2100 mg/kg (Rabbit)<br>OECD 402 | LC50 = 8.81 mg/L ( Rat ) 1 h |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

##### **(d) respiratory or skin sensitization;**

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

**(e) germ cell mutagenicity;** No data available

Not mutagenic in AMES Test

**(f) carcinogenicity;** No data available

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There are no known carcinogenic chemicals in this product

| Component          | EU | UK | Germany | IARC     |
|--------------------|----|----|---------|----------|
| Dimethyl acetamide |    |    |         | Group 2B |

(g) reproductive toxicity;  
Reproductive Effects No data available  
May cause harm to the unborn child.

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available  
Target Organs No information available.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects Do not empty into drains. .

| Component          | Freshwater Fish | Water Flea         | Freshwater Algae   | Microtox                                          |
|--------------------|-----------------|--------------------|--------------------|---------------------------------------------------|
| Dimethyl acetamide |                 | EC50 >500 mg/L/48h | EC50 >500 mg/L/72h | EC50 = 2393 mg/L 30 min<br>EC50 = 4815 mg/L 5 min |

Persistence and degradability  
Persistence Readily biodegradable  
Persistence is unlikely.

Bioaccumulative potential Bioaccumulation is unlikely

| Component          | log Pow | Bioconcentration factor (BCF) |
|--------------------|---------|-------------------------------|
| Dimethyl acetamide | 0.8     | No data available             |

Mobility in soil The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

Other adverse effects No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods  
Waste from Residues/Unused Waste is classified as hazardous Dispose of in accordance with the European Directives on

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|                               |                                                                                                                             |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Products</b>               | waste and hazardous waste Dispose of in accordance with local regulations                                                   |
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point.                                                   |
| <b>Other Information</b>      | Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains |

## SECTION 14: TRANSPORT INFORMATION

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>IMDG/IMO</b>                     | Not regulated                   |
| <b>Road and Rail Transport</b>      | Not regulated                   |
| <b>IATA</b>                         | Not regulated                   |
| <b>Special Precautions for User</b> | No special precautions required |

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** X = listed

| Component          | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|--------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Dimethyl acetamide | 204-826-4 | X    | X   | X     | X    | X    | X     | X    | KE-11114 |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%



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**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Prepared By

Health, Safety and Environmental Department

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Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**